**Name:** _____

**Date:** _____

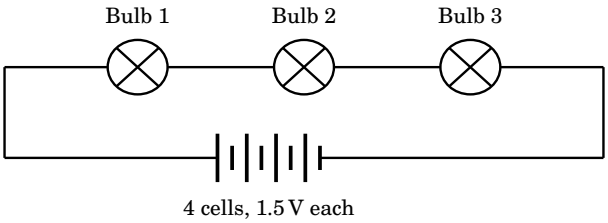
**Mr. Rawson • Y9 MYP Physics**

# 2.3 – Series Circuits: Build and Measure

*Unit 2: Electricity & Energy*

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**The question:** *when three bulbs share one loop, how is the battery's voltage shared out, and what happens to the current when you take a bulb away?*




**In a series circuit, the battery's voltage is SHARED between the bulbs.**

## Part 1 — Voltage in PhET

Open **Circuit Construction Kit: DC** (PhET).

1. Build a series circuit: **4 batteries and 3 bulbs** in one loop.
2. Click each battery and set its voltage to **1.5 V**.
3. Add a **voltmeter**. Put its two probes across **all 4 batteries**. Write the reading in the **PhET** column.
4. Move the probes across **Bulb 1**, then **Bulb 2**, then **Bulb 3**. Write each reading in the **PhET** column.

## Part 2 — Voltage in your real circuit

5. Build the same circuit with real cells and bulbs.
6. Put the voltmeter probes across **all 4 cells**. Write the reading in the **Real** column.
7. Move the probes across **each bulb** in turn. Write each reading in the **Real** column.
8. Add the three bulb voltages together. Write the total in the last row.

| **Voltage across …** | **PhET (V)** | **Real circuit (V)** |
|---|---|---|
| all 4 batteries (cells) | | |
| Bulb 1 | | |
| Bulb 2 | | |
| Bulb 3 | | |
| **Bulbs 1 + 2 + 3 added up** | | |

**Check**

In PhET, the three bulb voltages add up to the battery voltage **exactly**. In your real circuit they add up **close to** it. A small difference is normal, because the wires and the meter use up a little voltage themselves.

**1)** **Complete** the sentence.

The battery's voltage is _____ between the three bulbs, and the three bulb voltages _____ the battery voltage.

## Part 3 — Current: take a bulb out

9. Open the loop anywhere and put the **ammeter** in the gap, so the current goes *through* the meter. Read the current with **3 bulbs** and write down how bright the bulbs are.
10. **Predict** what the current will do when a bulb is taken out. Then take out **Bulb 3** and close the gap with a **lead**, so the loop is complete again. Read the ammeter and compare the brightness of the two bulbs left.
11. Take out **Bulb 2** and close the gap with a **lead**. Read the ammeter and compare the brightness of the last bulb.

Disconnect a lead after each reading.

| **Bulbs in the loop** | **My prediction** (circle one) | **Current (A)** | **Brightness** |
|---|---|---|---|
| 3 | — | | write how bright: |
| 2 | goes up / same / goes down | | brighter / same / dimmer |
| 1 | goes up / same / goes down | | brighter / same / dimmer |

**Brightness:** in the last two rows, circle the word that compares the bulbs with the row above.

**2)** **Explain** why the current rises each time a bulb is taken out.

**Some students — the voltage rule and the brightness**

The battery is **6 V** and the bulbs are identical. Use the voltage rule to find the voltage across **each** bulb with **3 bulbs**, with **2 bulbs** and with **1 bulb**. Then say what happens to the voltage across each bulb as bulbs are taken out.

3 bulbs: _____ 2 bulbs: _____ 1 bulb: _____